NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

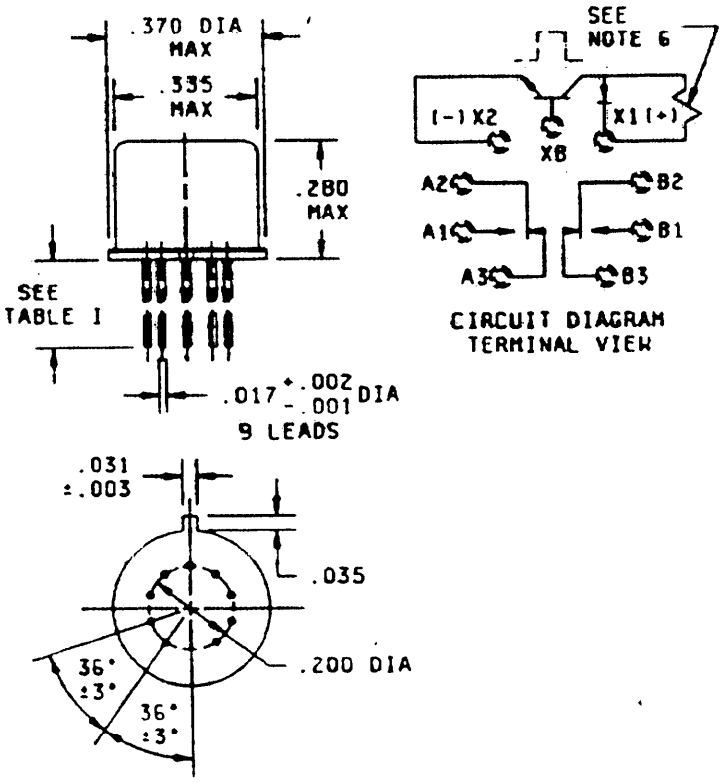
MIL-PRF-28776/1E
10 February 1984
SUPERSEDING
MIL-R-28776/1D
27 September 1982

PERFORMANCE SPECIFICATION SHEET

RELAYS, HYBRID, ESTABLISHED RELIABILITY, DPDT, LOW LEVEL TO 1.0 AMPERE (TRANSISTOR DRIVEN) (ELECTROMECHANICAL OUTPUT)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the relays described herein shall consist of this specification and the latest issue of MIL-R-28776.



Inches	mm	Inches	mm	Inches	mm
.001	0.03	.031	0.79	.335	8.51
.002	0.05	.035	0.89	.370	9.40
.003	0.08	.200	5.08		
.017	0.43	.280	7.11		

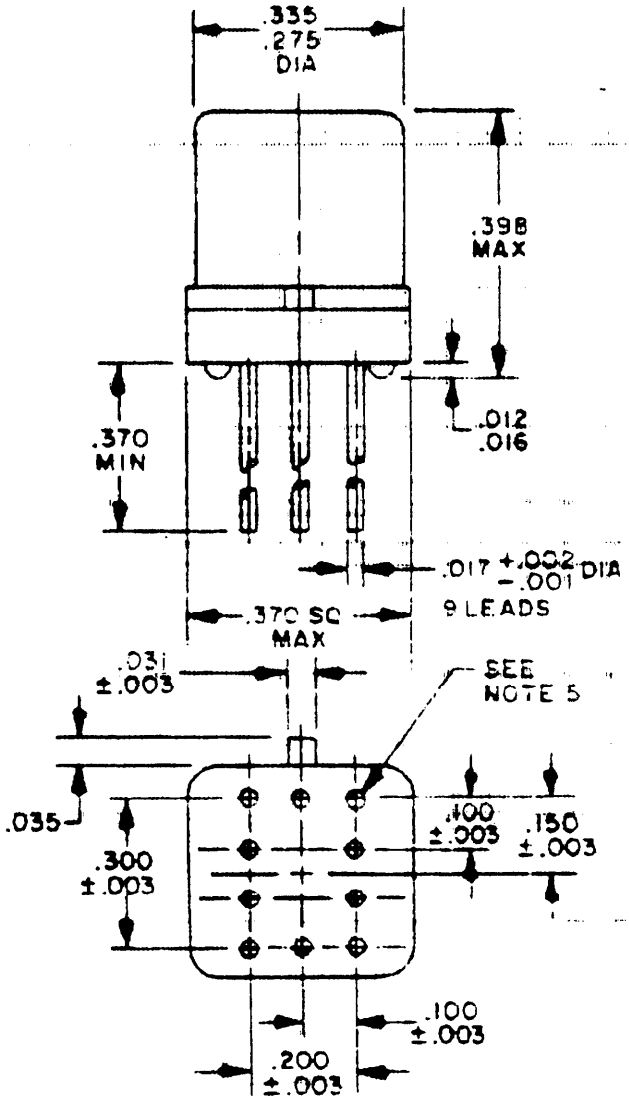
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm)
4. Terminal numbers shown above are for reference only. Numbers do not appear on relay.
5. Relays shall have a plus (+) sign placed on the circuit diagram as shown.
6. Coil symbol optional in accordance with MIL-STD-1285.
7. Circuit diagram shown on part is the terminal view.

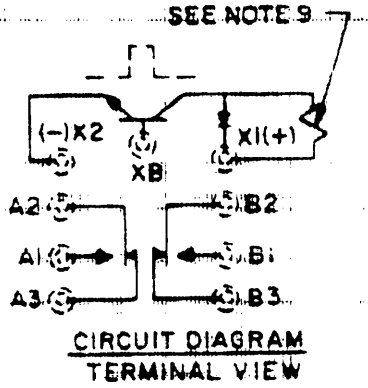
FIGURE 1. Dimensions and configuration

(6) denotes changes

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:09Z
Check the source to verify that this is the current version before use.





CIRCUIT DIAGRAM
TERMINAL VIEW

INCHES	MM
.012	0.30
.016	0.41
.035	0.89
.200	5.08
.275	6.99
.370	9.40
.398	10.11

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Spreader pads shall be centered to MIL-M-38510/75-08.
5. Dimensions and tolerances shown for the bottom view of the spreader pad are for the center to center locations of the holes in the spreader pad.
6. Shape optional within the envelope dimension.
7. Terminal numbers shown above for reference only. Numbers do not appear on relay.
8. Relays shall have a (+) sign placed on circuit diagram as shown.
9. Coil symbol optional in accordance with MIL-STD-1285.
10. Circuit diagram shown on part is the terminal view.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:09Z
Check the source to verify that this is the current version before use.

TABLE I. Dash numbers and characteristics. 1/

Dash numbers 2/				Supply voltage (V dc) 5/		At +25°C					Over temperature range			
Lead length			Spreader pad (fig 2) 4/			Coil resistance (Ohms) ±10%	Pickup voltage (V dc) Max	Hold voltage (V dc)	Dropout voltage (V dc)	Max base current to assure turn on 6/ (mA)	Pickup voltage (V dc) Max	Hold voltage (V dc)	Dropout voltage (V dc)	Max base current to assure turn on 6/ (mA)
1.500 min 3/	.187 ±.050, -.010	.500 min		Rated	Max									
013	019	025	031	5.0	5.8	50	2.7	1.4	0.22	.75	3.5	2.3	0.14	3.00
014	020	026	032	6.0	8.0	98	3.5	2.0	0.28	.55	4.5	3.2	0.18	2.04
015	021	027	033	9.0	12.0	220	5.3	3.0	0.54	.36	6.8	4.9	0.35	1.36
016	022	028	034	12.0	16.0	390	7.0	4.0	0.63	.27	9.0	6.5	0.41	1.03
017	023	029	035	18.0	24.0	880	10.5	6.0	0.91	.16	13.5	10.0	0.59	0.68
018	024	030	036	26.5	32.0	1,560	14.2	8.0	1.37	.13	18.0	13.0	0.89	0.50

1/ Each relay possesses high level and low level capabilities. However, relays previously tested or used above 50 mA resistive at 6 V dc maximum or peak ac open circuit are not recommended for subsequent use in low level applications.

2/ The suffix letter L, M, P, or R to designate the applicable failure rate level shall be added to the applicable listed dash number. Failure rate level (percent per 10,000 cycles): L, 3.0; M, 1.0; P, 0.1; R, 0.01. Example 013L - - - - - 024R.

3/ 1.500 leads are inactive for new design.

4/ Relays supplied with spreader pads (-031 through -036) shall have the pad rigidly attached.

5/ CAUTION: The use of any supply voltage less than the rated supply voltage will compromise the operation of the relay.

6/ Limit base current to 15 mA maximum.

MIL-R-83726/11

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:09Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

INPUT DATA:

Supply voltage: See table I.

Rated turn-off voltage (or current): 0.2 V dc maximum.

Base turn-on current: See table I.

Control signal voltage (or current): 15 milliamperes maximum.

Coil transient suppression: 1.0 volt maximum.

Turn-off time: 7.5 milliseconds maximum.

Turn-on time: 2 milliseconds maximum.

Voltage transients: Applicable.

Diode peak inverse voltage: 100 volts minimum.

CONTACT DATA:

Load ratings:

High level (relay case grounded):

(E) Resistive:
1.0 ampere at 28 V dc.
500 milliamperes at 115 V ac 400 Hz case not grounded.
250 milliamperes at 115 V ac 60 Hz case not grounded.
100 milliamperes at 115 V ac 60 and 400 Hz case grounded.
Inductive: 0.2 ampere at 28 V dc with 0.32 henry inductance.
(E) Lamp: 0.10 ampere at 28 V dc.

Low level: 10 to 50 microamperes at 10 to 50 millivolts dc or peak ac maximum.

Intermediate current: Applicable.

Contact resistance or voltage drop:

(E) Initial: 0.10 ohm maximum, 0.125 ohm maximum with spreader pad attached, (tested at 10 milliamperes maximum at 6 V dc maximum or peak ac).

High level:

During life: Not more than 5 percent of open circuit voltage.
(E) After life: 0.20 ohm maximum (0.225 ohm maximum with spreader pad attached).
During intermediate current: 1.0 ohm maximum.
(E) After intermediate current: 0.20 ohm maximum (0.225 ohm maximum with spreader pad attached).

Low level:

During life: 33 ohms maximum.
(E) After life: 0.15 ohm maximum (0.175 ohm maximum with spreader pad attached).

Contact bounce: 1.5 milliseconds maximum (applicable to failure rate level "L").
Contact stabilization time: 2.0 milliseconds maximum (applicable to failure rate levels "M", "P", and "R").

Overload (high level only): Two times rated current.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:09Z
Check the source to verify that this is the current version before use.

ELECTRICAL DATA:

Insulation resistance: 1/ 10,000 megohms at 500 V dc except the resistance between coil and case at high temperature shall be 1,000 megohms minimum.

Dielectric withstanding voltage: 1/

	Sea level V rms (60 Hz)	Altitude V rms (60 Hz)
Between case, frame, or enclosure and all contacts both in the energized and deenergized positions	500	125 - All terminals to case
Between case, frame, or enclosure and coil	500	
Between all contacts and coil	500	
Between open contacts in the energized and deenergized positions	500	
Between contact poles	500	

ENVIRONMENTAL DATA:

Temperature range: -65°C to +125°C.

Thermal shock: Applicable.

(E) Shock (specified pulse): MIL-STD-202, method 213, test condition B(75 G). Contact chatter shall not exceed 10 microseconds maximum (monitoring of open contacts not applicable).

(E) Vibration (sinusoidal): MIL-STD-202, method 204. Contact chatter shall not exceed 10 microseconds maximum (monitoring of open contacts not applicable).

Magnetic interference: Applicable.

Resistance to soldering heat: Applicable.

Acceleration: Applicable.

Salt atmosphere (corrosion): Method 1041, MIL-STD-750.

PHYSICAL DATA:

Terminal strength: Method 211, MIL-STD-202:

Pull test: Test condition A, 1-pound pull.
Bend test: Test condition C, 1/2-pound load.
Twist test: Test condition D.

Terminal solderability: Applicable.

Dimensions and configuration: See figures 1 and 2.

(E) Weight: 2.55 grams (0.09 ounce) maximum, 2.80 grams (0.099 ounce) maximum with spreader pad attached.

Seal: Hermetic.

Internal moisture: Applicable to failure rate levels M, P, and R.

Minimum marking: Military part number, "J" brand, date code, circuit diagram, and manufacturer's name or source code.

1/ Connect input leads together to avoid damage to the relay.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:09Z
Check the source to verify that this is the current version before use.

LIFE TEST REQUIREMENTS:

(E) High level: 100,000 cycles per relay.
(E) Low level: 100,000 cycles plus 900,000 mechanical life.

PART NUMBER: M28776/1-(dash number from table I and suffix letter designating failure rate level).

QUALIFICATION INSPECTION:

Qualification inspection and sample size: See table II.

TABLE II. Qualification inspection and sample size. 1/

Single submission	Group submission	
18 units plus 1 unsealed unit for level L at C = 0 2/ 33 units plus 1 unsealed unit for level M at C = 0 2/ Qualification inspection as applicable.	M28776/1-030	18 units plus 1 unsealed unit for level L at C = 0 2/ 33 units plus 1 unsealed unit for level M at C = 0 2/ Qualification inspection as applicable.
	M28776/1-025 M28776/1-026 M28776/1-027 M28776/1-028 M28776/1-029	3 units each part number Qualification inspection, group I.

1/ For retention of qualification or extension of qualification to lower failure rate levels, all life test data accumulated on MIL-R-28776/3 may be used in addition to MIL-R-28776/1 data.

2/ The number of units required for qualification testing will be increased as required in group IV, table II, MIL-R-28776, if the contractor elects to test the number of units permitting one or more failures.

(E) Initial qualification or relays supplied with spreader pads (-031 through -036) shall be tested as specified below:

Perform the following tests as specified in the qualification inspection table of MIL-R-39016, in the order shown below.

Before installation of pad: screening, visual and mechanical examination (internal), thermal shock, resistance to solvents, shock (specified pulse), acceleration, terminal strength, magnetic interference (when specified), capacitance (when specified), coil life (applicable to continuous duty relays only), resistance to soldering heat, salt spray (corrosion), overload (applicable to high level relays only), life, terminal strength, and intermediate current.

After installation of pad perform the following tests as specified in the qualification inspection table of MIL-R-39016, in the order shown below:

Insulation resistance, dielectric withstanding voltage, static contact resistance, pickup voltage, hold voltage, dropout voltage, coil resistance, operate and release time, contact dynamic characteristics, coil transient suppression (when specified), solderability, seal, visual and mechanical inspection (external).

(E) Qualification inspection (reduced testing for previously qualified relays) for relays supplied with spreader pads (-031 through -036), two (2) units of the 26.5 volt rated coil voltage, (-036) shall be tested as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:09Z
Check the source to verify that this is the current version before use.

Before installation of pad perform the following tests as specified in the qualification inspection table of MIL-R-39016 in the order shown below:

For failure rate level L only. Screening.
For failure rate levels M, P, and R: Vibration (sinusoidal) test duration shall be 10 minutes, particle impact noise detection (P.I.N.D., when specified), screening.

After installation of pad perform the following tests as specified in the qualification inspection table of MIL-R-39016 in the order shown below:

Insulation resistance, dielectric withstanding voltage, static contact resistance, pickup voltage, hold voltage, dropout voltage, coil resistance, operate and release time, contact dynamic characteristics, coil transient suppression (when specified), solderability, seal, visual and mechanical inspection (external).

(E) Group A testing for relays supplied with spreader pads (-031 through -036) shall be tested as specified below:

Before installation of pad perform subgroup 1 of group A tests.
After installation of pad perform subgroups 2 and 3 of group A tests.

Qualification inspection (reduced testing) and sample size: See table III. If the relays produced for MIL-R-28776/1 are similar in construction and design except for the coils to the relays produced for MIL-R-28776/3, then reduced testing for qualification of MIL-R-28776/1 relays may be performed concurrent with or subsequent to successful qualification of MIL-R-28776/3 relays.

TABLE III. Qualification inspection (reduced testing).

Examination or test
2 units each supply voltage - Group I of qualification inspection table.
1 unsealed sample unit - Internal examination

SUPERSESSION DATA:

Supersession data: See table IV.

Cross reference for Government logistical support: See table V.

TABLE IV. Supersession data. 1/

Superseded part number M28776/1-	New part number M28776/1-
001	013
002	014
003	015
004	016
005	017
006	018
007	019
008	020
009	021
010	022
011	023
012	024

1/ Dash numbers -013 through -018 are inactive for new design and are for support of existing equipment designs only.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:09Z
Check the source to verify that this is the current version before use.

TABLE V. Cross reference for Government logistical support.

Superseded part number M28776/1-	New part number M28776/1-	Support with part number M28776/1-	Superseded part number M28776/1-	New part number M28776/1-	Support with part number M28776/1-	New part number M28776/1-	Support with part number M28776/1-
001	013	013	010	022	028	030	030
002	014	014	011	023	029	031	031
003	015	015	012	024	030	032	032
004	016	016	---	025	025	033	033
005	017	017	---	026	026	034	034
006	018	018	---	027	027	035	035
007	019	025	---	028	028	036	036
008	020	026	---	029	029		
009	021	027	---	030	030		

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Air Force - 11, 17, 99
DLA - ES
NSA - NS

User activities:
Army - AR, ME
Navy - AS, OS, SH
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5945-0642-1)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:09Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-28776 / 1E	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Incl Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:09Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY
Naval Electronic Systems Command
Washington, DC 20363

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20363
ATTN: ELEX 8111

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:09Z
Check the source to verify that this is the current version before use.